

TOUQEER AHMAD

Data Scientist | Applied ML & Production AI Systems

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PROFESSIONAL SUMMARY

Data Scientist with hands-on experience applying unsupervised machine learning (K-Means, Hierarchical Clustering, DBSCAN) to large-scale real-world data (750K+ records), pursuing an M.Sc. in Data Science & AI at BITS Pilani. Distinguished by the ability to take models beyond the notebook: independently designs and ships production AWS systems (Lambda, DynamoDB, API Gateway, SES) that serve live users, giving equal fluency in model development and the engineering needed to deploy it. Civil engineering background adds a grounded, domain-driven approach to applied AI problems.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn), SQL, PostgreSQL, JavaScript/Node.js

Machine Learning: K-Means, Hierarchical (Ward/Average Linkage) & DBSCAN Clustering, Linear Regression, Random Forest, XGBoost, Feature Engineering, Silhouette Analysis, ANNs, Generative AI

Data Engineering: Kafka, PySpark, ETL Pipelines, Docker, Zookeeper

Analytics & Visualization: Excel (Advanced), Tableau, Matplotlib, Customer/Behavioral Segmentation

Cloud & Infrastructure: AWS (Lambda, DynamoDB, API Gateway, S3, CloudFront, Route 53, SES, SNS, ACM), Oracle Cloud Infrastructure, Oracle Cloud Generative AI

Software Engineering: Full-Stack Web Development, REST APIs, Serverless Architecture, Git/GitHub

KEY PROJECTS

Customer Behaviour Segmentation on Banking Transaction Data | *M.Sc. Apex Capstone Project* — 2025

- Applied K-Means, Hierarchical Clustering, and DBSCAN to segment ~753K bank customers by transaction behaviour, handling a heavily skewed dataset where 86.28% of customers had only a single transaction.
- Tuned and validated across techniques: K-Means (k=3) achieved a silhouette score of 0.3317 with balanced cluster sizes; identified severe chaining in average-linkage hierarchical clustering and established Ward linkage as the more suitable alternative; optimized DBSCAN to $\text{eps}=1.009$, $\text{min_samples}=3$.
- Delivered as a fully reproducible, runnable Colab notebook, correctly scoping evaluation methods to the unsupervised setting.

House Price Prediction Model | *M.Sc. Coursework, BITS Pilani* — 2026

- Trained and evaluated Linear Regression, Random Forest, and XGBoost models on Bengaluru housing data, applying feature engineering and hyperparameter tuning to improve predictive accuracy.

Streaming Data Ingestion Pipeline | *M.Sc. Coursework, BITS Pilani* — 2026

- Built a real-time streaming ingestion pipeline using Kafka, PySpark, PostgreSQL, Zookeeper, and Docker to process and land streaming data end-to-end.

Patient Booking & Authentication System | *Freelance | drtejasvisharma.in* — 2025 – Present

- Designed a full email-based OTP authentication and appointment booking system on AWS, and an SES bounce/complaint-handling pipeline (SNS → Lambda) that automatically flags suppressed patient emails in DynamoDB — direct experience turning a data signal into an automated, production decision.

Shaggykitchen — Serverless Food Ordering Platform | *Owner & Developer | shaggykitchen.in* — Feb 2026 – Present

- Shipped a production-grade serverless ordering platform (Customer, Owner, Delivery Agent roles) on AWS; built an Owner Analytics dashboard with churn tagging and menu performance metrics, and map-based delivery-fee logic using the Haversine formula — end-to-end ownership from data to decision to deployed feature.

Ecommerce Seller Recommendation System | *Personal Project* — 2025

- Built ETL pipelines across seller catalog, company sales, and competitor sales data in Python to recommend top-selling items.

Data Analytics & Data Science Job Simulations | *Forage — Deloitte Australia · Commonwealth Bank* — Feb 2026

- Completed real-world analytics tasks spanning Tableau dashboarding, forensic data analysis, exploratory data analysis, feature identification, and stakeholder communication of findings.

PROFESSIONAL EXPERIENCE

Data-Driven Client Advisory (Insurance) | *Star Health and Allied Insurance / LIC* — Aug 2023 – Present

- Segment client demographics and financial profiles using Excel-based analysis to tailor product recommendations and prioritize outreach — applied, informal analytics experience predating formal ML training.
- Track and analyze sales pipeline performance to identify conversion trends and inform retention strategy.

EDUCATION

M.Sc. in Data Science & Artificial Intelligence — **BITS Pilani** (Sep 2025 – Sep 2027, Expected)

B.E., Civil Engineering — Model Institute of Engineering and Technology (MIET), Jammu (2019 – 2022)

Diploma in Civil Engineering (3-Year) — Government Polytechnic College, Jammu (Sep 2016 – Jul 2019)

CERTIFICATIONS & TRAINING

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle (Oct 2025)
- Python Program — Coursera / UST (Sep 2025)
- Data Visualization — Coursera (Sep 2025)
- Excel & Master Data Analytics in Excel — Coursera / Great Learning (Sep 2025)
- Practical SQL Training — Great Learning (Sep 2025)

PROFESSIONAL MEMBERSHIPS

- Student Member, American Society of Civil Engineers (ASCE) — since Jul 2020